

Assignment 1

First order Model with Interaction Term (Quantitative Variables)

Deadline: November 2, 2025, by 11:59 pm. Submit to Gradescope Link in assignment 1 folder

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Problem 1:15 marks) The amount of water used by the production facilities of a plant varies. Observations on water usage and other, possibility related, variables were collected for 250 months. The dataset is given below.

```
water=read.csv("https://raw.githubusercontent.com/Thuntida/dataset/refs/heads/main/water.csv")
water
```

PROD <dbl>	TEMP <dbl>	HOUR <dbl>	USAGE <dbl>	DAYS <int>							
171.3	39.7	9.50	19.0	20							
19.4	16.0	20.00	6.6	21							
18.7	12.1	26.00	6.7	21							
25.6	39.0	24.00	9.5	21							
25.6	39.0	23.00	9.5	21							
139.2	14.3	16.00	12.2	21							
109.8	14.3	15.00	12.4	21							
206.9	8.4	14.00	12.9	21							
85.7	35.8	15.00	13.3	21							
172.5	18.1	15.00	14.4	21							
1-10 of 249 rows		Previous	1	2	3	4	5	6	...	25	Next

The explanatory variables are

TEMP= average monthly temperature (degree celsius)

PROD=amount of production (in hundreds of cubic)

DAYS=number of operationing day in the month (days)

HOUR=number of hours shut down for maintenance (hours)

The response variable is USAGE=monthly water usage (gallons/minute)

1.1 (2 marks) Fit the model containing all four independent variables. What is the estimated multiple regression equation?

```
full <- lm(USAGE ~ TEMP + PROD + DAYS + HOUR, data = water)
summary(full)
```

```
##
## Call:
## lm(formula = USAGE ~ TEMP + PROD + DAYS + HOUR, data = water)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -6.4030 -1.1433  0.0473  1.1677  5.3999
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  5.891627   1.028794   5.727 3.0e-08 ***
## TEMP         0.168673   0.008209  20.546 < 2e-16 ***
## PROD         0.040207   0.001629  24.681 < 2e-16 ***
## DAYS        -0.021623   0.032183  -0.672  0.502
## HOUR        -0.070990   0.016992  -4.178 4.1e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.768 on 244 degrees of freedom
## Multiple R-squared:  0.8885, Adjusted R-squared:  0.8867
## F-statistic:  486 on 4 and 244 DF,  p-value: < 2.2e-16
```

```
coef(full)
```

```
## (Intercept)      TEMP      PROD      DAYS      HOUR
##  5.89162697  0.16867306  0.04020739 -0.02162304 -0.07099009
```

Equation: $USAGE = 5.8916 + 0.1687TEMP + 0.04021PROD - 0.02162DAYS - 0.07099HOUR$ For every 1°C increase in TEMP, USAGE increases by 0.1687 gallons/min For every 1 unit increase in PROD, USAGE increases by 0.04021 gallons/min For every additional HOUR of shutdown, USAGE decreases by 0.07099 gallons/min DAYS has a small negative effect

1.2 (3 marks) Test the hypothesis for the full model i.e the test of overall significance. Use significance level 0.05. Ensure that you include the hypotheses, p-value, and conclusion. $H_0: \beta_{TEMP} = \beta_{PROD} = \beta_{DAYS} = \beta_{HOUR} = 0$ H_a : At least one β_j does not equal 0

```
anova(full) # ANOVA table
```

	Df <int>	Sum Sq <dbl>	Mean Sq <dbl>	F value <dbl>	Pr(>F) <dbl>
TEMP	1	2204.284528	2204.284528	704.8550829	6.530006e-74
PROD	1	3819.700985	3819.700985	1221.4102221	5.774726e-97
DAYS	1	1.092068	1.092068	0.3492063	5.551098e-01

	Df <int>	Sum Sq <dbl>	Mean Sq <dbl>	F value <dbl>	Pr(>F) <dbl>
HOUR	1	54.583131	54.583131	17.4538253	4.102173e-05
Residuals	244	763.058163	3.127288	NA	NA

5 rows

```
summary(full)$fstatistic
```

```
##      value      numdf      dendf
## 486.0171    4.0000 244.0000
```

$F(4,244) = 486.02$, $p\text{-value} < 2.2 \times 10^{-16}$ Since $p\text{-value} < 0.05$, we reject H_0 The model is statistically significant.

1.3 (2 marks) Would you suggest the model in part b for predictive purposes? Which model or set of models would you suggest for predictive purposes? Hint: Use Individual Coefficients Test (t-test) to find the best model. Make sure you provide your hypotheses, $p\text{-value}(s)$, and final model.)

```
summary(full)
```

```
##
## Call:
## lm(formula = USAGE ~ TEMP + PROD + DAYS + HOUR, data = water)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -6.4030 -1.1433  0.0473  1.1677  5.3999
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  5.891627   1.028794   5.727  3.0e-08 ***
## TEMP         0.168673   0.008209  20.546 < 2e-16 ***
## PROD         0.040207   0.001629  24.681 < 2e-16 ***
## DAYS        -0.021623   0.032183  -0.672  0.502
## HOUR        -0.070990   0.016992  -4.178  4.1e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.768 on 244 degrees of freedom
## Multiple R-squared:  0.8885, Adjusted R-squared:  0.8867
## F-statistic:  486 on 4 and 244 DF,  p-value: < 2.2e-16
```

```
reduced1 <- lm(USAGE ~ TEMP + PROD + DAYS, data = water)
summary(reduced1)
```

```
##
## Call:
## lm(formula = USAGE ~ TEMP + PROD + DAYS, data = water)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -7.0012 -1.1528  0.2578  1.2272  6.1756
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  3.812200    0.930119   4.099 5.65e-05 ***
## TEMP         0.181692    0.007846  23.158 < 2e-16 ***
## PROD         0.044427    0.001321  33.644 < 2e-16 ***
## DAYS        -0.019014    0.033240  -0.572   0.568
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.827 on 245 degrees of freedom
## Multiple R-squared:  0.8805, Adjusted R-squared:  0.879
## F-statistic: 601.8 on 3 and 245 DF,  p-value: < 2.2e-16
```

DAYS is not significant $H_0: \beta_{\text{DAYS}}=0$ $H_a: \beta_{\text{DAYS}} \neq 0$ $p=0.502 > 0.05 \rightarrow$ Fail to reject H_0 (remove DAYS) I would use this model: $\text{USAGE} = 3.8122 + 0.18169\text{TEMP} + 0.04443\text{PROD}$

1.4 (2 marks) Use Partial F test to confirm that the independent variable (removed from part c) should be out of the model at significance level 0.05. Ensure that you include the hypotheses, p-value, and conclusion.
 $H_0: \beta_{\text{DAYS}}=0$ (reduced model is okay) $H_a: \beta_{\text{DAYS}} \neq 0$

```
anova(reduced1, full)
```

	Res.Df <dbl>	RSS <dbl>	Df <dbl>	Sum of Sq <dbl>	F <dbl>	Pr(>F) <dbl>
1	245	817.6413	NA	NA	NA	NA
2	244	763.0582	1	54.58313	17.45383	4.102173e-05
2 rows						

$F=0.3492$, $p = 0.5551$ $p > 0.05$ so we Fail to reject H_0 So removing DAYS does not significantly reduce model fit (DAYS should be removed).

1.5 (2 marks) Obtain a 95% confidence interval of regression coefficient for TEMP from the model in part c. Give an interpretation.

```
confint(reduced1, "TEMP", level = 0.95)
```

```
##           2.5 %    97.5 %
## TEMP 0.1662385 0.197146
```

We are 95% confident that for each 1°C increase in TEMP, monthly water usage increases by 0.166 to 0.197 gallons/min, holding the amount of production (PROD) constant.

1.6 (2 marks) Use the method of Model Fit to calculate R_{adj}^2 and RSE to compare the full model in part a and the model in part c. Which model or set of models would you suggest for predictive purpose? For the final model, give an interpretation of R_{adj}^2 and RSE.

```
s_full <- sigma(full)
r2_full <- summary(full)$r.squared
adjr2_full <- summary(full)$adj.r.squared

# reduced
s_red <- sigma(reduced1)
r2_red <- summary(reduced1)$r.squared
adjr2_red <- summary(reduced1)$adj.r.squared

# print
data.frame(model = c("full", "reduced"),
           RSE = c(s_full, s_red),
           R2 = c(r2_full, r2_red),
           AdjR2 = c(adjr2_full, adjr2_red))
```

model <chr>	RSE <dbl>	R2 <dbl>	AdjR2 <dbl>
full	1.768414	0.8884861	0.8866580
reduced	1.826831	0.8805093	0.8790461
2 rows			

Full model explains slightly more variation as higher adjusted r-squared value. Full model has lower error (lower RSE).

So the full model performs better for prediction, and HOUR is significant (so we keep HOUR).

1.7 (2 marks) Build an interaction model to fit the multiple regression model from the model in part f. From the output, which model would you recommend for predictive purposes? Be sure to explain your process.

Model: USAGE~(TEMP+PROD+DAYS)^2

```
interact_all <- lm(USAGE ~ (TEMP + PROD + DAYS)^2, data = water)
summary(interact_all)
```

```
##
## Call:
## lm(formula = USAGE ~ (TEMP + PROD + DAYS)^2, data = water)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -6.2157 -0.5182  0.0781  0.5245  7.6784
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  7.805e+00  1.672e+00   4.668 5.05e-06 ***
## TEMP         7.640e-02  4.565e-02   1.674  0.0955 .
## PROD         1.263e-02  7.663e-03   1.648  0.1007
## DAYS        -3.698e-02  6.085e-02  -0.608  0.5439
## TEMP:PROD     1.032e-03  6.682e-05  15.448 < 2e-16 ***
## TEMP:DAYS    -1.790e-03  1.714e-03  -1.045  0.2972
## PROD:DAYS     2.760e-04  2.656e-04   1.039  0.2999
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.286 on 242 degrees of freedom
## Multiple R-squared:  0.9415, Adjusted R-squared:  0.9401
## F-statistic: 649.6 on 6 and 242 DF,  p-value: < 2.2e-16
```

```
coef(summary(interact_all))
```

```
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  7.8049186806 1.6720973346  4.6677418 5.046857e-06
## TEMP         0.0764018606 0.0456458230  1.6737974 9.546239e-02
## PROD         0.0126254968 0.0076625385  1.6476911 1.007138e-01
## DAYS        -0.0369845715 0.0608519590 -0.6077795 5.439037e-01
## TEMP:PROD     0.0010322439 0.0000668200  15.4481288 6.320787e-38
## TEMP:DAYS    -0.0017903493 0.0017136516 -1.0447569 2.971778e-01
## PROD:DAYS     0.0002759717 0.0002656231  1.0389597 2.998604e-01
```

```
anova(reduced1, interact_all)
```

	Res.Df <dbl>	RSS <dbl>	Df <dbl>	Sum of Sq <dbl>	F <dbl>	Pr(>F) <dbl>
1	245	817.6413	NA	NA	NA	NA
2	242	400.0081	3	417.6332	84.221	2.405576e-37
2 rows						

Only TEMP:PROD interaction is significant: p-value < 2e-16 Partial F-test comparing additive vs interaction: F = 84.22, p < 2.2e-16 So include the TEMP × PROD interaction → performance improves substantially.

USAGE=7.805+0.0764TEMP+0.0126PROD-0.00179TEMP·DAYS+0.001032TEMP·PROD+0.000276PROD·DAYS (TEMP:PROD is very significant so we keep main effects). The effect of temperature on water usage depends on production level. So when production increases, the slope relating TEMP to USAGE becomes steeper.

Problem 2:20 marks). A collector of antique grandfather clocks sold at auction believes that the price received for the clocks depends on both the age of the clocks and the number of bidders at the auction. Thus, (s)he hypothesizes the first-order model

$$y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \epsilon$$

where

y = Auction price (dollars)
 X_1 = Age of clock (years)
 X_2 = Number of bidders

A sample of 32 auction prices of grandfather clocks, along with their age and the number of bidders, is given in data file below

```
gfclocks=read.csv("https://raw.githubusercontent.com/Thuntida/dataset/refs/heads/main/GFCLOCKS.csv")
gfclocks
```

AGE <int>	NUMBIDS <int>	PRICE <int>
127	13	1235
115	12	1080
127	7	845
150	9	1522
156	6	1047
182	11	1979
156	12	1822
132	10	1253
137	9	1297
113	9	946

1-10 of 32 rows

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2.1 (2 marks) Use the method of least squares to estimate the unknown parameters β_0 , β_1 , β_2 of the model.

```
m2 <- lm(PRICE ~ AGE + NUMBIDS, data = gfclocks)
summary(m2)
```

```
##
## Call:
## lm(formula = PRICE ~ AGE + NUMBIDS, data = gfclocks)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -206.49 -117.34   16.66  102.55  213.50
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -1338.9513   173.8095  -7.704 1.71e-08 ***
## AGE          12.7406     0.9047   14.082 1.69e-14 ***
## NUMBIDS      85.9530     8.7285    9.847 9.34e-11 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 133.5 on 29 degrees of freedom
## Multiple R-squared:  0.8923, Adjusted R-squared:  0.8849
## F-statistic: 120.2 on 2 and 29 DF,  p-value: 9.216e-15
```

```
coef(m2)
```

```
## (Intercept)      AGE      NUMBIDS
## -1338.95134    12.74057    85.95298
```

PRICE = $-1338.95 + 12.74(\text{AGE}) + 85.95(\text{NUMBIDS})$ For each year of age, price increases by \$12.74, holding bidders constant. For each bidder, price increases by \$85.95, holding age constant.

2.2 (2 marks) Find the value of SSE that is minimized by the least squares method.

```
anova(m2)
```

	Df <int>	Sum Sq <dbl>	Mean Sq <dbl>	F value <dbl>	Pr(>F) <dbl>
AGE	1	2555224.5	2555224.48	143.40566	9.527372e-13
NUMBIDS	1	1727838.5	1727838.48	96.97066	9.344953e-11
Residuals	29	516726.5	17818.16	NA	NA
3 rows					

The value of SSE is 516727 that is minimized by the least squares method.

2.3 (2 marks) Estimate s , the standard deviation of the model, and interpret the result.

```
s <- sigma(m2)  # residual SE
s
```



```
## [1] 133.4847
```

Observed auction prices typically differ from model-predicted prices by about \$133.48.

2.4 (2 marks) Find and interpret the adjusted coefficient of determination, R^2_{Adj} .

```
adjR2 <- summary(m2)$adj.r.squared
adjR2
```

```
## [1] 0.8849194
```

About 88.49% of the variation in auction price is explained by age and number of bidders.

2.5 (4 marks) Construct the ANOVA table for the model and test the global F-test of the model at the $\alpha = 0.05$ level of significance. Be sure to state your hypotheses, p-values, provide the ANOVA table and make a conclusion.
 $H_0: \beta_1 = \beta_2 = 0$ (neither AGE nor NUMBIDS predicts PRICE) H_a : at least one β_j does not equal 0

```
anova(m2)
```

	Df <int>	Sum Sq <dbl>	Mean Sq <dbl>	F value <dbl>	Pr(>F) <dbl>
AGE	1	2555224.5	2555224.48	143.40566	9.527372e-13
NUMBIDS	1	1727838.5	1727838.48	96.97066	9.344953e-11
Residuals	29	516726.5	17818.16	NA	NA
3 rows					

```
summary(m2)
```

```
##
## Call:
## lm(formula = PRICE ~ AGE + NUMBIDS, data = gfclocks)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -206.49 -117.34   16.66  102.55  213.50
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -1338.9513   173.8095  -7.704 1.71e-08 ***
## AGE          12.7406     0.9047   14.082 1.69e-14 ***
## NUMBIDS       85.9530     8.7285    9.847 9.34e-11 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 133.5 on 29 degrees of freedom
## Multiple R-squared:  0.8923, Adjusted R-squared:  0.8849
## F-statistic: 120.2 on 2 and 29 DF,  p-value: 9.216e-15
```

Since $p < 0.05$, reject H_0 . The model is statistically significant.

2.6 (3 marks) Test the hypothesis that the mean auction price of a clock changes as the number of bidders increases when age is held constant (i.e., when $\beta_2 \neq 0$). (Use $\alpha = 0.05$) $H_0: \beta_2 = 0$ $H_a: \beta_2$ does not equal 0

```
coef(summary(m2))["NUMBIDS", c("Estimate", "Std. Error", "t value", "Pr(>|t|)")]
```

```
##      Estimate   Std. Error    t value    Pr(>|t|)
## 8.595298e+01 8.728523e+00 9.847368e+00 9.344953e-11
```

$p < 0.05$ so we reject H_0 . Number of bidders significantly affects price after accounting for age.

2.7 (2 marks) Find a 95% confidence interval for β_1 and interpret the result.

```
confint(m2, "AGE", level = 0.95)
```

```
##      2.5 %    97.5 %
## AGE 10.89017 14.59098
```

We are 95% confident that each extra year of age increases auction price by between \$10.89 and \$14.59, holding bidders constant.

2.8 (3 marks) Test the interaction term between the 2 variables at $\alpha = .05$. What model would you suggest to use for predicting y ? Explain.

```
m2_int <- lm(PRICE ~ AGE * NUMBIDS, data = gfclocks)
summary(m2_int)
```

```
##
## Call:
## lm(formula = PRICE ~ AGE * NUMBIDS, data = gfclocks)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -154.995  -70.431    2.069   47.880  202.259
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 320.4580   295.1413   1.086  0.28684
## AGE          0.8781     2.0322   0.432  0.66896
## NUMBIDS     -93.2648    29.8916  -3.120  0.00416 **
## AGE:NUMBIDS   1.2978     0.2123   6.112 1.35e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 88.91 on 28 degrees of freedom
## Multiple R-squared:  0.9539, Adjusted R-squared:  0.9489
## F-statistic: 193 on 3 and 28 DF, p-value: < 2.2e-16
```

```
coef(summary(m2_int))["AGE:NUMBIDS", c("Estimate", "Std. Error", "t value", "Pr(>|t|)")] # Test interaction coefficient
```

```
##      Estimate   Std. Error      t value      Pr(>|t|)
## 1.297846e+00 2.123326e-01 6.112325e+00 1.353474e-06
```

```
anova(m2, m2_int) # Compare additive vs interaction with partial F
```

	Res.Df <dbl>	RSS <dbl>	Df <dbl>	Sum of Sq <dbl>	F <dbl>	Pr(>F) <dbl>
1	29	516726.5	NA	NA	NA	NA
2	28	221362.1	1	295364.4	37.36052	1.353474e-06
2 rows						

p interaction = $1.35 \times 10^{-6} < 0.05$ The AGE $\times$ NUMBIDS interaction is statistically significant. Therefore, the interaction model should be used for prediction: PRICE ~ AGE * NUMBIDS The effect of age on price depends on number of bidders. _____

Problem 3:18 marks. Cooling method for gas turbines. Refer to the Journal of Engineering for Gas Turbines and Power (January 2005) study of a high pressure inlet fogging method for a gas turbine engine. The heat rate (kilojoules per kilowatt per hour) was measured for each in a sample of 67 gas turbines augmented with high pressure inlet fogging. In addition, several other variables were measured, including cycle speed (revolutions per minute), inlet temperature (degree Celsius), exhaust gas temperature (degree Celsius), cycle pressure ratio, and air mass flow rate (kilograms per second). The data are saved in the **TURBINE.CSV** file.

! [The first and last five observations are listed in the table.] (C:/Users- University of Calgary3.png)

```
turbine=read.csv("https://raw.githubusercontent.com/Thuntida/dataset/refs/heads/main/TURBINE.csv")
turbine
```

ENGINE <chr>	SHA... <int>	RPM <int>	CPRAT... <dbl>	INLET.TEMP <int>	EXH.TE... <int>	AIRFL... <int>	POW... <int>	HEATRA... <int>				
Traditional	1	27245	9.2	1134	602	7	1630	14622				
Traditional	1	14000	12.2	950	446	15	2726	13196				
Traditional	1	17384	14.8	1149	537	20	5247	11948				
Traditional	1	11085	11.8	1024	478	27	6726	11289				
Traditional	1	14045	13.2	1149	553	29	7726	11964				
Traditional	1	6211	15.7	1172	517	176	52600	10526				
Traditional	1	6210	17.4	1177	510	193	57500	10387				
Traditional	1	3600	13.5	1146	503	315	89600	10592				
Traditional	1	3000	15.1	1146	524	375	113700	10460				
Traditional	1	3000	15.0	1171	525	514	164300	10086				
1-10 of 67 rows				Previous	1	2	3	4	5	6	7	Next

3.1 (2 marks) Write a first-order model for heat rate (y) as a function of speed, inlet temperature, exhaust temperature, cycle pressure ratio, and air flow rate.

```
m3_full <- lm(HEATRATE ~ RPM + INLET.TEMP + EXH.TEMP + CPRATIO + AIRFLOW, data = turbine)
summary(m3_full)
```

```
##
## Call:
## lm(formula = HEATRATE ~ RPM + INLET.TEMP + EXH.TEMP + CPRATIO +
##     AIRFLOW, data = turbine)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1007.0  -290.9  -105.8   240.8  1414.0
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  1.361e+04  8.700e+02  15.649 < 2e-16 ***
## RPM          8.879e-02  1.391e-02   6.382 2.64e-08 ***
## INLET.TEMP  -9.201e+00  1.499e+00  -6.137 6.86e-08 ***
## EXH.TEMP     1.439e+01  3.461e+00   4.159 0.000102 ***
## CPRATIO      3.519e-01  2.956e+01   0.012 0.990539
## AIRFLOW     -8.480e-01  4.421e-01  -1.918 0.059800 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 458.8 on 61 degrees of freedom
## Multiple R-squared:  0.9235, Adjusted R-squared:  0.9172
## F-statistic: 147.3 on 5 and 61 DF,  p-value: < 2.2e-16
```

HEATRATE= $\beta_0 + \beta_1(\text{RPM}) + \beta_2(\text{INLET.TEMP}) + \beta_3(\text{EXH.TEMP}) + \beta_4(\text{CPRATIO}) + \beta_5(\text{AIRFLOW}) + \epsilon$
HEATRATE=13610+0.08879(RPM)−9.201(INLET.TEMP)+14.39(EXH.TEMP)+0.3519(CPRATIO)−0.8480(AIRFLOW)

3.2 (3 marks) Test the overall significance of the model using $\alpha = 0.01$. Be sure to state your hypotheses, p-values, and conclusion. $H_0: \beta_1 = \beta_2 = \beta_3 = \beta_4 = \beta_5 = 0$ H_a : At least one β_j does not equal 0 $F(5,61)=147.3$ p-value: $< 2.2 \times 10^{-16}$ so we reject H_0 . The regression model is statistically significant at $\alpha = 0.01$.

3.3 (2 marks) Fit the best additive model to the data using the method of least squares. Test significance of predictors at $\alpha = 0.06$. Be sure to state your hypotheses, p-values, and conclusion. CPRATIO p-value>0.06 so we will remove. We will keep the other variables as p-value<0.06.

Best model: HEATRATE~RPM+INLET.TEMP+EXH.TEMP+AIRFLOW

```
m3_add <- lm(HEATRATE ~ RPM + INLET.TEMP + EXH.TEMP + AIRFLOW, data=turbine)
summary(m3_add)
```

```
##
## Call:
## lm(formula = HEATRATE ~ RPM + INLET.TEMP + EXH.TEMP + AIRFLOW,
##     data = turbine)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1007.7  -290.5  -106.0   240.1  1414.8
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  1.362e+04  8.133e+02  16.744 < 2e-16 ***
## RPM          8.882e-02  1.344e-02   6.608 1.02e-08 ***
## INLET.TEMP  -9.186e+00  7.704e-01 -11.923 < 2e-16 ***
## EXH.TEMP     1.436e+01  2.260e+00   6.356 2.76e-08 ***
## AIRFLOW     -8.475e-01  4.370e-01  -1.939  0.057 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 455.1 on 62 degrees of freedom
## Multiple R-squared:  0.9235, Adjusted R-squared:  0.9186
## F-statistic: 187.1 on 4 and 62 DF,  p-value: < 2.2e-16
```

3.4 (3 marks) Test all possible interaction terms for the best model in part (c) at $\alpha = .06$. What is the final model would you suggest to use for predicting y ? Explain.

```
m3_int <- lm(HEATRATE ~ (RPM + INLET.TEMP + EXH.TEMP + AIRFLOW)^2, data=turbine)
anova(m3_add, m3_int)
```

	Res.Df <dbl>	RSS <dbl>	Df <dbl>	Sum of Sq <dbl>	F <dbl>	Pr(>F) <dbl>
1	62	12841965	NA	NA	NA	NA
2	56	8717640	6	4124325	4.41561	0.001010171
2 rows						

H_0 : No interaction terms are needed H_a : At least one interaction term improves the model $F = 4.4156$, p -value = 0.00101 Since $p < 0.06$ we reject H_0 . So model: $\text{HEATRATE} \sim (\text{RPM} + \text{INLET.TEMP} + \text{EXH.TEMP} + \text{AIRFLOW})^2$

3.5 (2 marks) Give practical interpretations of the β_i estimates. additive model: $\text{HEATRATE} \sim \text{RPM} + \text{INLET.TEMP} + \text{EXH.TEMP} + \text{AIRFLOW}$ For every increase of 1 RPM, heat rate increases by about 0.0888 kJ/kW/hr, keeping other variables constant. For every 1°C increase in inlet temperature, heat rate decreases by about 9.19 kJ/kW/hr, holding others constant. An large inlet temperature improves the efficiency. For every 1°C increase in exhaust temperature, heat rate increases by about 14.36 kJ/kW/hr → worse efficiency. For every 1 kg/s increase in air flow, heat rate decreases by about 0.85 kJ/kW/hr (slight efficiency improvement).

3.6 (2 marks) Find RSE, s from the model in part (d) Residual standard error: 455.1 The typical prediction error is about 455.1 kJ/kW/hr.

3.7 (2 marks) Find the adjusted-R² value from the model in part (d) and interpret it. Adjusted R-squared: 0.9186
91.86% of the variation in heat rate is explained by the model. This means the model fits the data very well.

3.8 (2 marks) Predict a heat rate (y) when a cycle of speed = 273,145 revolutions per minute, inlet temperature= 1240 degree celsius, exhaust temperature=920 degree celsius, cycle pressure ratio=10 kilograms persecond, and air flow rate=25 kilograms persecond.

```
newdata <- data.frame(RPM = 273145,  
                      INLET.TEMP = 1240,  
                      EXH.TEMP = 920,  
                      AIRFLOW = 25)  
predict(m3_add, newdata)
```

```
##          1  
## 39682.04
```

The model predicts the heat rate will be about 39680 kJ/kW/hr.
